

MLVICX: Multi-Level Variance-Covariance Exploration for Chest X-Ray Self-Supervised Representation Learning

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Abstract—Self-supervised learning (SSL) reduces the need for manual annotation in deep learning models for medical image analysis. By learning the representations from unlabelled data, self-supervised models perform well on tasks that require little to no fine-tuning. However, for medical images, like chest X-rays, characterised by complex anatomical structures and diverse clinical conditions, a need arises for representation learning techniques that encode fine-grained details while preserving the broader contextual information. In this context, we introduce MLVICX (Multi-Level Variance-Covariance Exploration for Chest X-ray Self-Supervised Representation Learning), an approach to capture rich representations in the form of embeddings from chest X-ray images. Central to our approach is a novel multi-level variance and covariance exploration strategy that effectively enables the model to detect diagnostically meaningful patterns while reducing redundancy. MLVICX promotes the retention of critical medical insights by adapting global and local contextual details and enhancing the variance and covariance of the learned embeddings. We demonstrate the performance of MLVICX in advancing self-supervised chest X-ray representation learning through comprehensive experiments. The performance enhancements we observe across various downstream tasks highlight the significance of the proposed approach in enhancing the utility of chest X-ray embeddings for precision medical diagnosis and comprehensive image analysis. For pertaining, we used the NIH-Chest X-ray dataset. Downstream tasks utilized NIH-Chest X-ray, Vinbig-CXR, RSNA pneumonia, and SIIM-ACR Pneumothorax datasets. Overall, we observe up to 3% performance gain over SOTA SSL approaches in various downstream tasks. Additionally, to demonstrate generalizability of our method, we conducted additional experiments on fundus images and observed superior performance on multiple datasets. Codes are available at GitHub.

Index Terms—Self-supervised learning, medical image analysis, chest X-Ray, representation learning,

variance-covariance alignment, multi-level alignment, clinical diagnosis.

I. INTRODUCTION

MEDICAL image analysis is a critical component of modern healthcare, enabling clinicians to diagnose diseases accurately, plan treatment strategies, and monitor patient progress. Traditionally, it requires laborious manual identification and analysis of features within images, leading to subjectivity and time constraints. The advent of deep learning and AI address these challenges through automated feature extraction from medical images, ultimately aiding the diagnosis process [1], [2]. While deep learning models have demonstrated remarkable potential in various medical image analysis tasks, their performance is governed by the quality and diversity of the labeled data used for training. Acquiring accurate labels for medical images is a labor-intensive task that demands specialized expertise, accurate annotation, and significant time investment [3], [4].

Self-supervised learning (SSL) [5], [6], [7], [8] mitigate the challenges of acquiring labeled data due to its ability to learn representations with limited annotated data [9], [10], [11]. It involves a two-step training process: pre-training and downstream training. During pre-training, a model is trained using pretext tasks designed to generate supervisory signals from the unlabeled data. As the model solves these tasks, it captures useful patterns, structures, and relationships in the data, effectively learning informative features without requiring external annotations. The learned representations are fine-tuned for the downstream tasks, using a smaller labeled dataset to adapt the pre-trained features to a specific task. Many studies have shown the effectiveness of SSL in various medical image analysis tasks, including classification [12], [13], [14], detection [15], and segmentation [16], [17], [18].

Recent advancements in SSL leveraged discriminative techniques in which the model extracts generalized features by distinguishing between different variations of the same input. Discrimination SSL falls into two main categories: contrastive and non-contrastive methods, which widely adopt the joint-embedding framework, often realized through Siamese network architecture [19]. Contrastive SSL methods bring embeddings of positive pairs closer together while pushing them away from embeddings of negative pairs. The positive pairs are the

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augmented versions of the same image, while negative pairs consist of augmented versions of different images [20]. On the other side, non-contrastive methods bring together embeddings of positive pairs but avoid the use of explicit negative pairs, either by choosing specific architectural design choices or by adopting explicit regularization techniques [21].

Nonetheless, conventional SSL [5], [7], [22], [23] techniques have been adapted and applied to medical image analysis tasks. However, certain limitations emerge that hinder the quality of learned representations while effectively navigating the complexities inherent to medical images, particularly chest X-rays. One notable challenge is effectively fusing coarse and fine-grained details within chest X-rays. Chest X-rays encompass a wide spectrum of information, ranging from subtle anomalies like pulmonary nodules or infiltrates to anatomical features such as the heart silhouette or lung contours. For instance, in the case of pulmonary nodules, an accurate diagnosis requires the localization of small, subtle lesions within the lung tissue. These anomalies are often challenging to detect due to their size and location within the chest cavity. On the other hand, anatomical features like the heart silhouette or diaphragm shape are essential for understanding the overall clinical context of the patient's condition. Analyzing the heart silhouette can provide valuable information about the heart's size, shape, and position within the chest, which is essential for diagnosing heart conditions and assessing overall cardiac health. Meanwhile, examining lung contours can help detect abnormalities, such as masses or fluid accumulation within the lungs, and can also provide information about the lung's overall condition and position in the chest. Achieving a balance between capturing these subtle anomalies and preserving the broader anatomical context is critical to effective chest X-ray analysis. [24], [25]

Most existing SSL methods successfully capture either global or fine-grained features; however, they often struggle to integrate both [26], [27]. Global features offer a broad understanding of the entire image, encompassing its overall structure and context. The fine-grained features signify anomalies or abnormalities, offering detailed insights into localized areas of interest. This disparity in feature extraction leads to representations that either lack the broader global context or overlook the fine-grained details [28], [29], [30].

To address these limitations, we propose MLVICX (Multi-Level Variance-Covariance Exploration for Chest X-ray Self-Supervised Representation Learning). MLVICX captures intermediate feature maps from diverse layers of its twinned backbone encoders. These features pass through context-bottleneck blocks [29] to extract and enhance hierarchical contextual information, resulting in locally guided global representations. The hierarchical information aggregates as local contextual features, encouraging the model to learn fine-grained representations. MLVICX also applies explicit variance and covariance regularization on global and local contextual features to enhance feature variability in the representational space. Our contributions are as follows:

- 1) We introduce MLVICX to leverage the global and local contextual feature maps from various intermediate levels of the twinned backbone encoders.

- 2) We extract context information from local intermediate feature maps through context-bottleneck blocks while concurrently enhancing the features captured by the backbone encoders for locally guided global representation.
- 3) We implement multi-level variance-covariance exploration to regulate global and local contextual details, enhancing feature variability at various levels.

While previous approaches [31], [32], [33] have focused on extracting both global and local features, they often overlook representation variability and add computational complexity by incorporating reconstruction modules. In contrast, MLVICX introduces multi-level variance-covariance exploration to regulate global and local contextual details. Through explicit regulation of features across multiple intermediate levels using variance and covariance terms, MLVICX enhances variability at various levels and adapts the utilization of global and local contextual details. This yields an informative and inherently discriminative representation space. We assess the representations acquired through the proposed approach by comparing them against SOTA self-supervised methods, including SimCLR [5], MoCo [6], BYOL [7], VICReg [23], and Barlow-Twins [34]. Additionally, within the domain of chest X-ray analysis, we compare our approach with the recently proposed Inter_MoCo [35], PCRLv1 [32], and PCRLv2 [31]. Our comprehensive evaluation demonstrates a noteworthy enhancement in performance across all considered datasets, highlighting the effectiveness of the representations learned by the proposed approach.

II. RELATED WORKS

A. Self Supervised Learning

SSL offers various pretraining options, where one approach involves designing hand-crafted pretext tasks [36], [37], [38], such as predicting image rotations [39] or contextually consistent patches, solving a jigsaw puzzle [40], image colorization [41] etc. Another possible approach is instance discrimination methods, wherein contrastive methods like SimCLR [5], MoCo [6] etc., non-contrastive methods such as BYOL [7], SimSiam [8], Barlow-Twins (BT) [34], VICReg [23] etc. are employed. These methods are designed to foster the acquisition of invariant image representations, thereby mitigating the impact of image transformations. Contrastive methods aim to bring embeddings of similar instances (positive pairs) closer while pushing apart embeddings of dissimilar instances (negative pairs). Non-contrastive methods also emphasize similarity but avoid explicit negative pairs and can be further categorized into self-distillation-based approaches like BYOL [7] and SimSiam [8] and information maximization methods such as Barlow-Twins [34] and recently proposed VICReg [23]. Another line of work in SSL involves clustering and generative methods where clustering-based [42], [43], [44] tasks encourage models to group similar instances together, while generative tasks [45], [46] involve generating parts of an image and learning to reconstruct the entire image from these parts.

B. SSL for Chest X-Ray Images

Given the inherent scarcity of precisely annotated medical data, SSL is a compelling approach in this domain. For instance, [47] proposed a semi-supervised framework for medical image segmentation that used pseudo-label guided contrastive learning to enhance inter-class separability and intra-class compactness. CTNet [48] is another recent method utilizing a contrastive Transformer backbone and multiscale and high-resolution feature modules for polyp segmentation in colonoscopy images. Another interesting method is SAM [49] based upon pixel-level contrastive SSL for generating anatomical embeddings in CT scans. Among its various methodologies, instance discrimination techniques, particularly contrastive approaches, have gained considerable attention and applicability in chest X-ray images. For instance, Hari et al. introduced MoCo-CXR [12], an adaptation of the MoCo [6] framework tailored for harnessing meaningful representations from chest X-ray images. Drawing inspiration from SimCLR [5], Azizi et al. presented, Multi-Instance Contrastive Learning (MICLe) in [14]. This method leverages the power of multiple patient images to construct positive pairs, resulting in better representations compared to SimCLR. Similarly, MedAug [13] introduced a novel strategy for selecting positive pairs across varying views using metadata. An interesting work, DiRA [50], introduced by Haghighi et al., unites discriminative, restorative, and adversarial learning to collaboratively learn complementary visual information from unlabeled medical images for fine-grained semantic representation learning. Zhou et al. introduced preservational contrastive representation learning (PCRL) for both X-ray and CT modalities [32]. This method enhances medical representations by dynamically reconstructing diverse image contexts. Subsequently, the authors presented PCRLv2 in [31], a refined version of PCRL that unifies pixel-level restoration and scale information, yielding better results across various tasks and datasets. Kaku et al. incorporate intermediate-layer closeness, extending beyond relying on the final layer, which led to superior feature quality compared to standard MoCo [35]. While these approaches showed benefits, they commonly overlooked the explicit handling of representation variability.

In the context of chest X-ray analysis, representation variability is important due to the commonality of anatomical structures across patients. Representation variability refers to the diversity of representations captured by the model. It ensures that the model learns to recognize and differentiate between the diverse appearances of anatomical structures, such as the heart, lungs, and ribs, along with the subtle abnormalities. For instance, subtle abnormalities or pathologies may manifest differently in different patients across different anatomical regions, and the model needs to be able to distinguish these variations to make accurate diagnoses. Without it, the model might generalize too broadly while missing the variations present in real-world data. In contrast, MLVICX is uniquely tailored to address representational variability effectively. It leverages a multi-level framework to enhance variance and covariance progressively, resulting in refined feature maps from diverse layers. This approach captures fine-grained local details and overarching semantic concepts.

III. METHODOLOGY

Fig. 1 provides the overview of the proposed SSL framework, which is based on a siamese configuration of a deep neural network. The following subsections provide details on the particulars of MLVICX.

A. Proposed

SSL Framework As shown in Fig. 1, the proposed SSL framework employs a joint embedding architecture that incorporates two branches, each consisting of a backbone CNN encoder (f_θ and $f_{\theta'}$ respectively) followed by global MLP heads (g_ϕ and $g_{\phi'}$) and multi-level MLP heads (h_ϕ and $h_{\phi'}$). The input image, $x \in \mathcal{D}$ where $\mathcal{D}$ is the training set, undergoes random image transformations t and t' to generate two perturbed variants v and v' . These transformations encompass spatial-level operations and pixel-level adjustments. The spatial transformations include flipping, rotation, and random cropping. These transformations alter the spatial arrangement or structure of the image while preserving the semantic information. The pixel-level transformations include Gaussian blur, color jittering, and solarization. The views v and v' are passed through f_θ and $f_{\theta'}$ to generate global representations, denoted as $y = f_\theta(v)$ and $y' = f_{\theta'}(v')$ respectively. Global representations refer to the abstract, high-level features of the entire image learned by the model during training.

1) Intermediate Representations and Context-Bottleneck Heads: MLVICX extracts intermediate representations to encode contextual information from varying layers of complexity. These intermediate representations are pivotal in integrating global and local contextual details within chest X-rays. The encoders (f_θ and $f_{\theta'}$) generate intermediate feature maps at different levels of abstraction, referred to as $y_p \in \{y_0, y_1, \dots, y_n\}$ and $y'_p \in \{y'_0, y'_1, \dots, y'_n\}$ respectively, where p corresponds to the p th layer of the encoder. The dimensions of y_p and y'_p are denoted as $c_p \times h_p \times w_p$, reflecting the number of channels c_p along with the height h_p and width w_p . These intermediate feature maps y_p and y'_p are used by context-bottleneck heads (C_{t_p} and C'_{t_p}) respectively. Here, C_{t_p} represents a particular instance at the p th layer, while there are n such heads.

A context-bottleneck head consists of the channel pooling block and the spatial pooling block. The channel pooling block takes y_p as input and produces a 1D channel-activated map $a_p^c \in \mathbb{R}^{c_p \times 1 \times 1}$ according to the formulation:

$$a_p^{ch} = \sigma(\text{Conv}_1(\text{AvgPool}(y_p)) + \text{Conv}_1(\text{MaxPool}(y_p))) \quad (1)$$

Here, σ represents the sigmoid activation function. The channel-activated map a_p^{ch} is then modulated with y_p using element-wise multiplication to yield the feature map $k_p^{ch} = (a_p^{ch} \otimes y_p) \in \mathbb{R}^{c_p \times h_p \times w_p}$. Additionally, Conv_1 refers to a convolutional layer with a 1×1 kernel size. The spatial pooling block is complementary to the channel pooling block. It takes the feature map k_p^c as input and generates a 2D spatially-activated map $a_p^s \in \mathbb{R}^{1 \times h_p \times w_p}$ using the formulation:

$$a_p^s = \sigma(\text{Conv}_7([\text{AvgPool}(k_p^{ch}); \text{MaxPool}(k_p^{ch})])) \quad (2)$$

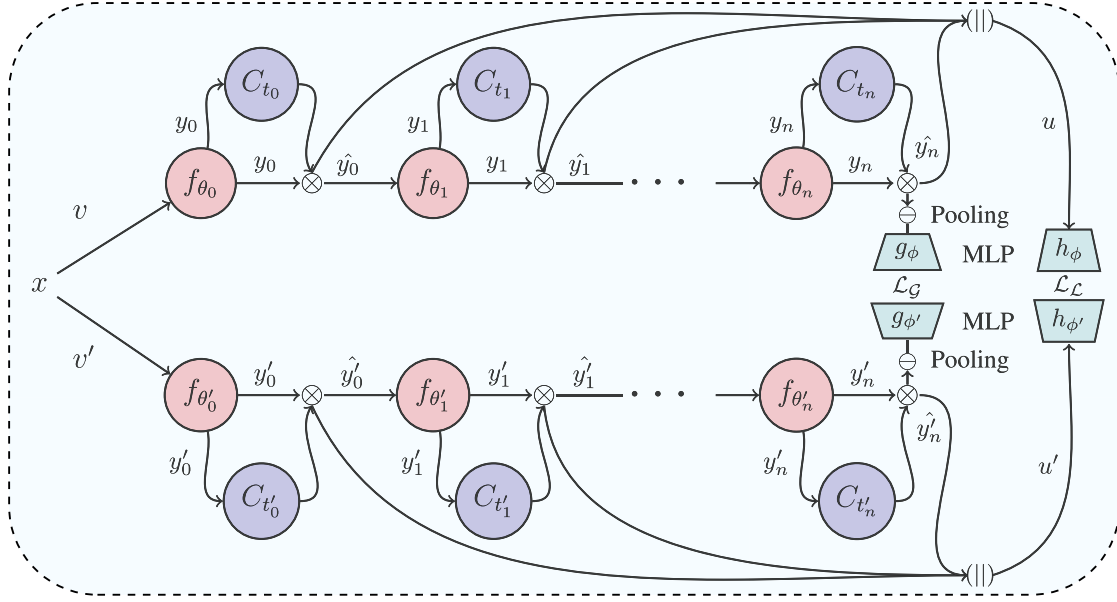


Fig. 1. Architecture of the proposed approach. v and v' are two augmented versions of the input x , processed by siamese encoders f_θ and $f_{\theta'}$ respectively to give y and y' . C_{t_p} and $C_{t'_p}$ are the context-bottleneck blocks to refine the intermediate representations y_p and y'_p . The refined intermediate representations are aggregated to make compound representations u and u' in each branch. g_ϕ and $g_{\phi'}$ are global MLP heads and h_ϕ and $h_{\phi'}$ are multi-level MLP heads. These heads serve as expanders to project the embeddings into higher-dimensional space. $(||)$ represents the concatenation of multi-level intermediate feature maps, while $\otimes$ represents multiplication.

Similarly, the spatially-activated map a_p^s is modulated with k_p^{ch} using element-wise multiplication to yield the refined feature map $\hat{y}_p = (a_p^s \otimes k_p^{ch})$. The process described above is symmetrically applied to obtain $\hat{y}'_p$, the refined feature map from the other branch. The refined intermediate representations $\hat{y}_p$ and $\hat{y}'_p$ are processed by subsequent layers of the encoder, which enables the model to learn locally guided context-aware global representations y and y' . Subsequently, for the multi-level variance-covariance exploration, we combine $\hat{y}_p$ and $\hat{y}'_p$ to construct compound feature representations u and u' both residing in the space $\mathbb{R}^{c \times h \times w}$. The aggregation process is a critical step that blends the information extracted from diverse intermediate feature maps. The aggregated compound feature representations u and u' encapsulate an amalgamation of information extracted from various layers, incorporating both local and global contextual cues. Before the aggregation occurs, a pre-processing step is applied to ensure consistency in terms of h and w across different $\hat{y}_p$ and $\hat{y}'_p$.

B. Loss Function

The MLP heads (g_ϕ and $g_{\phi'}$) along with (h_ϕ and $h_{\phi'}$) serve as expansion components, projecting the global representations (y and y') and the aggregated compound representations (u and u') into higher-dimensional spaces respectively. This enables the subsequent hierarchical computation of loss functions. Specifically, g_ϕ and $g_{\phi'}$ transform y and y' into global embeddings denoted as $z = g_\phi(y)$ and $z' = g_{\phi'}(y')$, while h_ϕ and $h_{\phi'}$ map the compound representations into $m = h_\phi(u)$ and $m' = h_{\phi'}(u')$. Motivated from [23], we defined the following two losses for global embeddings z and z' and multi-level embeddings m and m' :

$$\mathcal{L}_G = \alpha(inv(Z, Z')) + \beta(var(Z) + var(Z')) + \gamma(cov(Z) + cov(Z')) \quad (3)$$

$$\mathcal{L}_L = \alpha(inv(M, M')) + \beta(var(M) + var(M')) + \gamma(cov(M) + cov(M')) \quad (4)$$

where, Z and M represent batches of global embeddings (z and m), while following a similar analogy Z' and M' represent batches of corresponding multi-level embeddings (z' and m').

The overall loss function for MLVICX is a composite of $\mathcal{L}_G$ and $\mathcal{L}_L$, as outlined in (5).

$$\mathcal{L}_{mlca} = \mathcal{L}_G + \lambda \times \mathcal{L}_L \quad (5)$$

Here, λ is the balance factor. The symbols inv , var and cov are defined according to (6), (7) and (8) respectively.

$$inv(Z, Z') = \frac{1}{b} \sum_{i=1}^b \|z_i - z'_i\|_2^2 \quad (6)$$

The (6) calculates the Euclidean distance between each pair of corresponding representations (z_i and z'_i) in the batches Z and Z' . This quantifies how different the representations learned by the two branches are on average. The $var(Z)$ term encourages the representations in batch Z to have a variance for each dimension above a threshold Λ , which is set as 1.

$$var(Z) = \frac{1}{r} \sum_{j=1}^r \max \left(0, \Lambda - \left(\sqrt{Var(z_{:,j})} + \epsilon \right) \right) \quad (7)$$

$$Var(z_{:,j}) = \frac{1}{b-1} \sum_{i=1}^b (z_{ij} - \bar{z}_j)^2, \quad \bar{z}_j = \frac{1}{b} \sum_{i=1}^b z_{ij}$$

Here, b refers to batch size, and r represents the dimensionality of the feature vector. The hyperparameters Λ and ϵ are introduced to govern the standard deviation's target value and ensure numerical stability, respectively. The $cov(Z)$ term captures the covariance between different dimensions of a batch of representations Z by taking the square of the off-diagonal components of the covariance matrix $C(Z)$. The off-diagonal elements of this matrix represent the covariances between different dimensions, indicating how these dimensions vary together across data points. Minimizing $cov(Z)$ encourages the representations to have low covariances between their dimensions, which can help to capture more independent and discriminative features.

$$C(Z) = \frac{1}{b-1} \sum_i^b (z_{i:} - \bar{z})(z_{i:} - \bar{z})^T, \bar{z} = \frac{1}{b} \sum_i^b z_{i:}$$

$$cov(Z) = \frac{1}{r} \sum_{\ell \neq q} C(Z)_{\ell q}^2 \quad (8)$$

The ultimate aim of these regularization terms is to enhance the qualities of the embeddings: $inv(Z, Z')$ measures their similarity, $var(Z)$ enforces variability across the batch, and $cov(Z)$ emphasizes decorrelation between different dimensions of the embeddings. The coefficients α , β , and γ are scalar factors that weigh the contributions of the different loss terms. Analogous to Z , the inv , var , and cov terms are applied to M to enhance the quality of the multi-level embeddings. $inv(M, M')$ measures the similarity between corresponding multi-level embeddings m_i and m'_i in batches M and M' , respectively. Similarly, $var(M)$ enforces variability control across the batch, and $cov(M)$ encourages decorrelation between different dimensions of the multi-level embeddings.

IV. EXPERIMENTAL SETUP

In this section, we detail our experiments and provide a description of the datasets, followed by the evaluation protocols, implementation details, and baselines.

A. Descriptions of Datasets

NIH Chest X-ray-14 Dataset: In our pretraining phase, we use the NIH-Chest X-ray-14 dataset [51]. This dataset encompasses a vast collection of 112,120 X-ray images procured from 30,805 distinct subjects. Each image comes with one or more of 14 labels, including thorax-related diseases and a no-finding category. We use the official 86,524 training samples, resized to dimensions of 224×224 during the pretraining phase, while 25,596 test samples are used for subsequent evaluations in downstream tasks.

Vinbig-CXR Dataset: The Vinbig-CXR dataset [52] contains 18K postero-anterior view X-ray scans with annotations for localization and classification of common thoracic disease. The samples are officially divided into 15k for training and 3k for testing. In our study, we use the Vinbig-CXR dataset to conduct the classification tasks downstream evaluations.

SIIM-ACR Pneumothorax Dataset: The SIIM-ACR [53] dataset is a benchmark for detecting pneumothorax, a condition characterized by air in the pleural cavity leading to lung collapse. Comprising a total of 12,047 chest radiographs, each paired with manually created segmentation masks specifically targeting pneumothorax. We carefully curated an equal number of positive (pneumothorax) and negative (healthy) samples for our evaluations. For segmentation tasks, We randomly allocated 20% of the samples for validation, containing both positive and negative, which is a standard practice.

RSNA Pneumonia: The RSNA [54] dataset consists of 30k chest radiographs captured from a frontal view, each annotated as either healthy or pneumonia-positive. This dataset is employed to assess the adaptability of our pre-trained representations for classification tasks across external datasets.

B. Evaluation Protocols

To assess the effectiveness of the learned representations from our proposed approach, we conducted downstream tasks involving chest X-ray image classification and segmentation. These tasks were executed using only the backbone encoder obtained from pretraining.

Following established practices in the literature such as SimCLR [5], BYOL [7], PCRLv2 [31], and VICReg [23], we adopted two distinct evaluation protocols to showcase the robustness and efficiency of our framework: 1) fine-tuning and 2) frozen. For classification tasks, we added a single linear layer classifier in both evaluation settings on top of the pre-trained backbone CNN encoder. In the fine-tuning evaluation protocol, we fine-tuned both the SSL pretrained backbone CNN encoder and the newly added linear classifier layer. Conversely, in the frozen evaluation protocol, the parameters of the backbone CNN encoder remain fixed, ensuring that only the linear layer's parameters are updated during downstream training. We presented results for downstream training on the test/validation dataset using various subsets (1%, 10%, 30%, and All) of the training data.

For the segmentation tasks, we adopted a U-Net architecture. We initialized the U-Net encoder's weights with the SSL pre-trained encoder's weights. Under the fine-tuning setting, both the backbone encoder and the decoder are updated during downstream training, while under the frozen evaluation setting, only the decoder's weights are updated in downstream training. Notably, the U-Net decoder is initialized randomly. In both evaluation settings, we evaluated the representations learned by our proposed SSL method using different subsets (30% and 100%) of the downstream training data. The 1% and 10% subsets are omitted due to limited data availability. To effectively measure the performance of our method, we utilized the AUC score for classification tasks and the Dice score for segmentation tasks.

C. Implementation Details

MLVICX employs the ResNet18 architecture as the backbone encoder for the SSL pretraining phase. This training process encompasses a batch size of 64 over a duration of 300 epochs. Within the MLVICX framework, the expander heads consist

of three linear layers, each configured with a dimensionality of 2048. These layers are augmented with ReLU activation functions and batch normalization. Our optimization strategy relies on the LARS (Layer-wise Adaptive Rate Scaling) algorithm, utilizing a learning rate of $3e-5$ and applying a weight decay of $1e-5$. The hyperparameters α , β , and γ are directly adopted from VICReg [23]. During downstream evaluations, we maintain a batch size of 64 for classification tasks on the NIH dataset. For the Vinbig-CXR and RSNA datasets, a batch size of 16 is used. For the classification tasks on the NIH dataset, we fine-tune with a learning rate of $1e-5$. In contrast, for the Vinbig-CXR and RSNA datasets, the learning rate is adjusted to $1e-3$. For segmentation tasks across all datasets, we set the batch size to 32, and the learning rate is fine-tuned to $3e-4$ for optimal performance. In all downstream tasks, we incorporate early stopping with patience of 10 epochs, ensuring that the training process stops efficiently based on validation performance criteria.

D. Baselines

Our comparative analysis thoroughly evaluated the proposed approach against various baseline methods encompassing supervised and SSL techniques across various categories. We trained the model using labeled data with our designated settings for supervised training from scratch (Rand. Init.). In the SSL domain, we included SOTA contrastive and non-contrastive methods, SimCLR [5], MoCov2 [6], BYOL [7], SimSiam [8], Barlow-Twins (BT) [34], and VICReg [23]. We conducted the pretraining for these baselines ourselves, following their official implementations and keeping in alignment with our proposed approach's training protocol. Specifically, we conducted the pretraining for 300 epochs using a batch size of 64 without modifying their default hyperparameters or network configurations. For more relevant comparison, we also incorporated SSL baselines proposed for chest X-ray analysis, namely Inter_MocCo [35], PCRLv1 [32], PCRLv2 [31] and CAiD [33]. We utilized the official publicly available implementations of these methods. The pretraining is conducted on the official training dataset provided by the NIH, as mentioned in the previous section.

V. RESULTS AND DISCUSSION

Table I presents the evaluation test set results of our proposed framework MLVICX, along with considered baselines, on the downstream classification task using the NIH dataset. These results represent the average of three independent runs, where SD values are not displayed due to their small magnitudes. This Table highlights the potential of SSL techniques to extract meaningful features from chest X-ray images, even when only a limited amount of labeled data is accessible for fine-tuning. The Table illustrates the performance of different methods when their pre-trained representations are fine-tuned on subsets of the training data (1%, 10%, 30%, and 100%) using the linear evaluation protocol. The reported metrics include the mean AUC and class-wise AUC scores for different medical conditions.

Observing the results, we can infer that MLVICX consistently outperforms the baselines across different subsets of training

data samples, considered for fine-tuning. Particularly, for the 1% data subset, the proposed approach achieves an AUC of 0.642, which is a notable improvement over the closest competing baseline with an AUC of 0.630. Notably, it achieves the highest mean AUC score among all methods, indicating its superior ability to extract meaningful representations. In the 10% labeled data scenario, the proposed method consistently achieves the highest overall AUC scores with a mean AUC of 0.733. For the 30% subset, MLVICX achieved AUC of 0.763, which is 3% higher than the closest baseline PCRLv2. A similar trend is observed for the 100% dataset. Note that PCRLv2 is a dedicated SSL approach for medical data. Further, our method also exhibits remarkable class-wise improvements across various medical conditions.

Table II comprehensively evaluates the MLVICX model's generalization capabilities on out-of-distribution medical imaging datasets, specifically the Vinbig-CXR and RSNA datasets. We evaluated the model's performance under frozen and fine-tuning settings within the linear evaluation protocol, with results presented on the test sets regarding the mean AUC score. We also assessed the performance of MLVICX on the NIH dataset, employing the frozen evaluation setting. MLVICX consistently performed well across all the considered datasets in this setting. Notably, with only 1% of labeled data from NIH, it achieves an AUC score of 0.601, surpassing all considered baselines. This demonstrates the model's ability to effectively utilize pre-trained knowledge, making it suitable for scenarios with limited labeled data. Furthermore, MLVICX consistently outperformed other methods as the data percentage increased. For the Vinbig-CXR dataset, MLVICX maintains an advantage over its competitors in the frozen setting, with AUC scores consistently higher across different data percentages. Particularly, with only 1% of the data, MLVICX achieved an AUC score of 0.709, showcasing its generalization ability. On the RSNA dataset, MLVICX maintains a better performance in the frozen setting as well. Similarly, for the fine-tuning setting, MLVICX outperformed other methods across different data percentages on the Vinbig-CXR and RSNA datasets. Impressively, with 30% of the data for fine-tuning on Vinbig-CXR, MLVICX achieved an AUC score of 0.927. In low-data scenarios, such as fine-tuning with only 1% of the data, MLVICX maintained its effectiveness, achieving an AUC score of 0.815 on RSNA.

While MLVICX demonstrates superior performance across multiple datasets and evaluation settings, there are few instances where its performance is slightly lower, particularly with out-of-distribution datasets like Vinbig-CXR and RSNA. However, the differences in these cases are marginal; for instance, with 1% labeled samples from RSNA, the best AUC score is 0.790, while that of MLVICX is 0.783. Similarly, with 10% labeled samples from Vinbig-CXR, the performance margin between MLVICX and PCRLv2 is only 0.0003 in the AUC score. With small fractions of the data, the sample size is extremely limited, making it highly sensitive to inherent data variability. The slightly lower AUC of MLVICX could be attributed to the variability in small sample sizes rather than any fundamental limitation of the model. Despite these slight dips, MLVICX remains highly competitive, demonstrating its robustness and effectiveness even in

TABLE I

EVALUATION OF THE PROPOSED METHOD ALONGSIDE VARIOUS BASELINES ON THE DOWNSTREAM CLASSIFICATION TASK USING THE NIH DATASET

Data %	Methods	Atelectasis	Cardiomegaly	Consolidation	Edema	Effusion	Emphysema	Fibrosis	Hernia	Infiltration	Mass	Nodule	Pleural_Thick.	Pneumonia	Pneumothorax	Mean
1%	Rand. Init	0.570	0.627	0.616	0.677	0.647	0.478	0.648	0.533	0.611	0.517	0.487	0.528	0.585	0.513	0.577
	SimCLR [5]	0.650	0.647	0.567	0.658	0.715	0.504	0.663	0.646	0.627	0.592	0.548	0.623	0.580	0.632	0.621
	MoCov2 [55]	0.605	0.618	0.619	0.651	0.665	0.535	0.644	0.556	0.614	0.572	0.565	0.583	0.565	0.605	0.602
	BYOL [7]	0.609	0.631	0.621	0.657	0.670	0.543	<u>0.690</u>	0.583	0.594	0.567	0.620	0.609	0.570	0.589	0.610
	SimSiam [8]	0.610	0.611	0.601	0.641	0.647	0.524	0.670	0.603	0.579	0.572	0.589	0.619	0.569	0.608	0.603
	Barlow-Twins [34]	0.619	0.678	0.620	0.677	0.672	0.464	0.664	0.668	0.624	0.568	0.501	0.601	0.578	0.604	0.614
	VicReg [23]	0.629	0.652	0.539	0.689	0.668	0.467	0.646	0.609	0.627	0.583	0.507	0.576	0.573	0.562	0.600
	Inter_MoCo [35]	0.620	0.580	0.613	0.608	0.699	0.532	0.538	0.521	0.625	0.570	0.526	0.590	0.543	0.630	0.601
	PCRLv1 [32]	0.627	<u>0.690</u>	0.589	0.693	0.704	0.523	0.690	0.451	0.628	<u>0.615</u>	0.552	0.622	0.582	0.626	0.617
	CAiD [33]	0.595	0.565	<u>0.650</u>	0.614	0.699	0.480	0.612	0.476	0.649	0.573	0.508	0.592	0.572	<u>0.646</u>	0.594
	PCRLv2 [31]	<u>0.640</u>	0.693	0.607	0.761	<u>0.706</u>	0.596	0.623	0.438	0.617	0.645	<u>0.575</u>	<u>0.622</u>	<u>0.595</u>	0.673	<u>0.630</u>
	MLVICX	0.656	0.632	0.657	<u>0.735</u>	0.719	<u>0.553</u>	0.667	<u>0.625</u>	<u>0.638</u>	0.556	0.596	0.638	0.598	0.635	0.642
10%	Rand. Init	0.650	0.595	0.671	0.761	0.692	0.609	0.703	0.770	0.624	0.601	0.586	0.628	0.604	0.691	0.656
	SimCLR [5]	0.675	0.772	0.690	0.756	0.714	0.646	0.707	0.782	0.640	0.672	0.575	0.667	0.629	0.722	0.689
	MoCov2 [55]	0.665	0.702	0.640	0.766	0.719	0.686	0.657	0.748	0.661	0.681	0.601	0.697	0.646	0.702	0.684
	BYOL [7]	0.635	0.732	0.643	0.726	0.701	0.648	0.679	0.723	0.672	0.623	0.654	<u>0.689</u>	0.669	0.682	0.677
	SimSiam [8]	0.668	0.679	0.693	0.765	0.718	0.651	0.721	0.773	0.633	0.593	0.541	0.627	0.632	0.688	0.668
	Barlow-Twins [34]	0.680	0.765	0.691	0.776	0.741	0.675	0.689	0.702	0.711	0.641	0.656	0.617	<u>0.658</u>	0.635	0.718
	VicReg [23]	0.671	0.690	0.685	0.776	0.722	0.636	0.726	<u>0.795</u>	0.660	0.628	0.621	0.643	0.631	0.696	0.684
	Inter_MoCo [35]	0.705	<u>0.767</u>	0.705	<u>0.804</u>	0.770	0.680	0.702	0.597	0.662	0.704	0.622	0.674	0.626	0.727	0.696
	PCRLv1 [32]	0.669	<u>0.702</u>	0.694	0.785	0.734	0.635	0.702	0.751	0.664	0.646	0.575	0.657	0.630	0.705	0.682
	CAiD [33]	<u>0.701</u>	0.527	0.693	0.740	0.760	0.650	0.647	0.540	0.665	0.637	0.605	0.656	0.588	0.696	0.653
	PCRLv2 [31]	0.708	0.764	0.713	0.801	<u>0.785</u>	<u>0.760</u>	<u>0.735</u>	0.682	<u>0.676</u>	0.699	<u>0.670</u>	<u>0.689</u>	0.654	<u>0.768</u>	<u>0.720</u>
	MLVICX	0.689	0.757	<u>0.708</u>	0.806	0.788	0.762	0.762	0.835	0.677	0.695	0.675	0.700	0.651	0.783	0.733
30%	Rand. Init	0.694	0.791	0.687	0.778	0.751	0.707	0.715	0.740	0.654	0.674	0.610	0.691	0.650	0.749	0.707
	SimCLR [5]	0.710	0.825	0.701	0.793	0.781	0.723	0.733	0.728	0.666	0.685	0.642	0.709	0.653	0.773	0.722
	MoCov2 [55]	0.704	0.808	0.693	0.801	0.705	0.728	0.749	0.775	0.673	<u>0.731</u>	0.649	0.705	0.655	0.774	0.719
	BYOL [7]	0.698	0.791	0.691	0.797	0.759	0.679	0.743	0.807	0.666	0.669	0.630	0.695	0.642	0.746	0.716
	SimSiam [8]	0.690	0.783	0.688	0.793	0.764	0.672	0.745	0.832	0.650	0.672	0.640	0.685	0.643	0.746	0.713
	Barlow-Twins [34]	0.701	0.835	0.697	0.791	0.776	0.695	0.737	0.758	0.661	0.703	0.624	0.700	<u>0.659</u>	0.750	0.719
	VicReg [23]	0.702	0.807	0.687	0.809	0.765	0.712	<u>0.747</u>	0.779	0.639	0.691	0.626	0.701	0.657	0.762	0.718
	Inter_MoCo [35]	0.701	0.821	<u>0.708</u>	0.798	0.773	0.739	0.734	0.794	0.651	0.730	0.658	0.693	0.637	<u>0.776</u>	0.727
	PCRLv1 [32]	0.689	<u>0.827</u>	0.681	0.793	0.768	0.702	0.744	0.752	0.670	0.687	0.640	0.697	0.645	0.754	0.716
	CAiD [33]	<u>0.721</u>	0.806	0.685	0.790	<u>0.786</u>	0.729	0.732	<u>0.835</u>	0.656	0.710	<u>0.666</u>	0.700	0.647	0.768	0.728
	PCRLv2 [31]	0.714	0.822	0.699	<u>0.804</u>	0.782	0.722	<u>0.747</u>	0.809	<u>0.675</u>	<u>0.714</u>	0.641	0.714	0.648	0.768	<u>0.731</u>
	MLVICX	0.730	0.859	0.709	0.816	0.795	0.821	0.785	0.843	0.682	0.754	0.691	0.733	0.692	0.822	0.763
All	Rand. Init	0.721	0.851	0.709	0.816	0.792	0.740	0.751	0.782	0.684	0.741	0.657	0.724	0.652	0.779	0.741
	SimCLR [5]	0.731	0.866	0.724	0.828	0.799	0.781	0.767	0.869	0.679	0.730	0.627	0.735	0.672	0.808	0.756
	MoCov2 [55]	0.723	0.839	0.736	0.809	0.791	0.749	0.751	0.851	0.689	0.734	0.649	0.731	0.668	0.798	0.751
	BYOL [7]	0.728	0.863	0.721	0.824	0.796	0.734	0.760	0.826	0.672	0.731	0.669	0.727	0.664	0.788	0.748
	SimSiam [8]	0.728	0.857	0.717	0.825	0.797	0.742	0.752	0.789	0.683	0.727	0.640	0.725	0.677	0.794	0.744
	Barlow-Twins [34]	0.728	0.861	0.724	0.825	0.792	0.771	0.752	0.841	0.669	0.736	0.649	0.717	0.670	0.785	0.749
	VicReg [23]	0.730	0.864	0.721	0.823	0.800	0.775	0.760	0.804	0.678	0.750	0.667	0.732	0.684	0.799	0.754
	Inter_MoCo [35]	0.735	0.864	0.725	0.832	0.808	0.814	<u>0.782</u>	0.848	0.674	0.766	0.677	<u>0.740</u>	0.679	0.811	0.765
	PCRLv1 [32]	0.733	0.861	0.720	0.822	0.801	0.795	<u>0.782</u>	0.848	0.661	0.745	0.668	0.728	0.670	0.806	0.758
	CAiD [33]	<u>0.751</u>	<u>0.867</u>	<u>0.730</u>	<u>0.828</u>	<u>0.818</u>	<u>0.824</u>	<u>0.776</u>	<u>0.850</u>	<u>0.696</u>	<u>0.773</u>	<u>0.707</u>	0.735	<u>0.708</u>	<u>0.826</u>	<u>0.774</u>
	PCRLv2 [31]	0.730	0.863	0.720	0.814	0.802	0.814	0.777	0.842	0.684	0.748	0.682	0.731	0.687	0.799	0.761
	MLVICX	0.759	0.884	0.736	0.837	0.821	0.886	0.809	0.870	0.702	0.788	0.717	0.758	0.709	0.854	0.790

The table presents the test set results after three independent runs under the finetuning linear evaluation protocol on different subsets of training data (1%, 10%, 30%, and 100%), measured in terms of mean auc scores. Class-wise AUC scores are reported for each medical condition, demonstrating MLVICX's consistent improvements across a diverse range of pathologies compared to other self-supervised and supervised baselines. Bold font indicates the best performance, and underlined text represents the second-best performance.

low-data scenarios. These findings highlight MLVICX's adaptability and consistency in medical image classification tasks. It effectively handles diverse data distributions, demonstrating better AUC scores in frozen and fine-tuning settings. This shows its potential suitability for practical clinical applications with limited datasets.

Furthermore, we also measure the performance of the proposed SSL framework for segmentation in the downstream task. Table III presents the performance of MLVICX and considered baselines in the context of transfer learning for segmentation tasks, specifically focusing on pneumothorax segmentation within the SIIM-ACR dataset under both frozen and finetuning settings. The evaluation metric used here is the Dice coefficient,

which measures the similarity between predicted and ground truth segmentations. Remarkably, MLVICX consistently outperforms all other methods on both the 30% subset of the dataset and the entire dataset. Specifically for the frozen scenario in which models are evaluated directly after pertaining, MLVICX shows comparatively better performance, demonstrating the significance of performance improvement achieved by MLVICX.

In the fine-tuning setting, where pre-trained models undergo additional training on the specific pneumothorax segmentation task, MLVICX's performance is consistently better. It achieves the highest dice coefficients for both the 30% dataset subset and the entire dataset, with approximately 2% of performance gain. In these findings, it becomes evident that MLVICX excels not

TABLE II

PERFORMANCE EVALUATION OF MLVICX AND THE CONSIDERED BASELINES ON NIH, VINBIG-CXR, AND RSNA DATASETS UNDER BOTH FROZEN AND FINE-TUNING SETTINGS

Methods	Data %	Rand. Init	SimCLR [5]	MoCov2 [55]	BYOL [7]	SimSiam [8]	BT [34]	VicReg [23]	Inter_MoCo [35]	PCRLv1 [32]	CAiD [33]	PCRLv2 [31]	MLVICX
NIH (Frozen)	1%	0.516	0.569	0.561	0.547	0.539	0.584	0.587	0.524	0.566	0.567	0.579	0.601
	10%	0.571	0.627	0.621	0.617	0.611	0.642	0.627	<u>0.659</u>	0.620	0.654	0.643	0.661
	30%	0.611	0.676	0.669	0.663	0.659	0.660	0.662	<u>0.689</u>	0.655	0.683	0.675	0.691
	ALL	0.618	0.700	0.689	0.690	0.688	0.680	0.673	0.703	0.672	<u>0.705</u>	0.691	0.710
Vinbig-CXR (Frozen)	1%	0.568	0.683	0.643	0.656	0.650	0.661	0.653	0.686	0.695	<u>0.698</u>	0.598	0.709
	10%	0.594	0.774	0.736	0.778	0.775	0.787	0.765	0.804	0.772	0.781	0.719	<u>0.793</u>
	30%	0.670	0.826	0.813	0.821	0.816	0.827	0.814	<u>0.829</u>	0.814	0.824	0.790	0.832
	ALL	0.711	0.853	0.855	0.840	0.841	0.844	0.832	<u>0.859</u>	0.850	0.853	0.851	0.861
RSNA (Frozen)	1%	0.757	0.769	0.768	0.765	0.762	0.780	0.767	0.769	0.768	0.790	0.768	0.783
	10%	0.768	0.805	0.801	0.784	0.778	0.808	0.804	0.812	0.805	<u>0.810</u>	0.803	0.812
	30%	0.770	0.814	0.808	0.801	0.798	0.811	0.809	<u>0.815</u>	0.802	0.814	0.805	0.819
	ALL	0.782	0.818	0.812	0.809	0.805	0.817	0.812	0.820	<u>0.811</u>	0.820	<u>0.811</u>	0.820
Vinbig-CXR (Finetune)	1%	0.722	0.779	0.771	0.762	0.758	0.769	0.754	0.761	0.776	<u>0.781</u>	0.759	0.808
	10%	0.847	0.877	0.862	0.858	0.859	0.875	0.865	0.894	0.873	0.891	0.903	0.898
	30%	0.883	0.899	0.890	0.889	0.871	0.903	0.890	<u>0.919</u>	0.892	0.919	0.916	0.927
	ALL	0.901	0.924	0.920	0.917	0.915	0.919	0.919	0.930	0.914	<u>0.938</u>	<u>0.940</u>	0.948
RSNA (Finetune)	1%	0.770	0.800	0.794	0.783	0.781	0.800	0.795	0.804	0.789	<u>0.810</u>	0.801	0.815
	10%	0.798	0.811	0.813	0.805	0.801	0.809	0.807	<u>0.820</u>	0.799	0.818	0.815	0.824
	30%	0.800	0.818	0.820	0.814	0.816	0.820	0.817	<u>0.828</u>	0.813	0.822	0.823	0.832
	ALL	0.810	<u>0.830</u>	0.827	0.825	0.821	0.829	0.827	<u>0.830</u>	0.820	<u>0.830</u>	0.829	0.840

Results are reported for different data percentages. The results for the NIH and Vinbig-CXR datasets are reported in terms of AUC, while accuracy is used as the evaluation metric for the RSNA dataset. These results represent the average performance across three independent runs, and SD is not shown due to minimal variability. Bold represents the best performance, while underlined text represents the second best.

TABLE III

EVALUATION OF SEMANTIC-SEGMENTATION SIIM-ACR PNEUMOTHORAX UNDER BOTH FROZEN AND FINETUNING SETTINGS OF SSL EVALUATION

Methods	SIIM-ACR Pneumothorax (Dice)			
	Frozen		Finetune	
	30%	All	30%	All
Rand. Init	0.42 ± 0.00	0.48 ± 0.01	0.45 ± 0.04	0.51 ± 0.01
SimCLR [5]	0.44 ± 0.03	0.50 ± 0.01	0.46 ± 0.00	0.53 ± 0.03
MoCov2 [55]	0.43 ± 0.02	0.50 ± 0.02	0.46 ± 0.01	0.52 ± 0.02
BYOL [7]	0.43 ± 0.02	0.49 ± 0.01	0.45 ± 0.04	0.52 ± 0.01
SimSiam [8]	0.43 ± 0.02	0.49 ± 0.01	0.45 ± 0.04	0.52 ± 0.02
Barlow-Twins [34]	0.44 ± 0.03	0.48 ± 0.03	0.46 ± 0.02	0.54 ± 0.03
VicReg [23]	0.43 ± 0.01	0.48 ± 0.02	0.46 ± 0.02	0.54 ± 0.04
Inter_MoCo [35]	0.44 ± 0.02	0.53 ± 0.01	0.47 ± 0.02	0.55 ± 0.01
PCRLv1 [32]	0.43 ± 0.02	0.48 ± 0.01	0.46 ± 0.02	0.52 ± 0.00
CAiD [33]	0.44 ± 0.02	0.55 ± 0.01	0.46 ± 0.01	0.56 ± 0.01
PCRLv2 [31]	0.44 ± 0.01	0.56 ± 0.00	0.46 ± 0.04	<u>0.57 ± 0.02</u>
MLVICX	0.45 ± 0.01	0.57 ± 0.00	0.48 ± 0.01	0.59 ± 0.01

We reported test set results for 30% and 100% data. The metric used is the dice score. The best performance is bold, while underlined text represents the second best.

TABLE IV

PERFORMANCE EVALUATION OF THE PROPOSED APPROACH IN TERMS OF AUC SCORE ON THE MUReD AND ODIR DATASETS

Methods	MuReD		ODIR	
	Frozen	Finetune	Frozen	Finetune
	10%	10%	10%	10%
Rand. Init	0.586	0.667	0.564	0.632
SimCLR [5]	0.721	0.809	0.702	0.734
MoCov2 [55]	0.733	0.812	0.714	0.740
BYOL [7]	0.705	0.786	0.674	0.713
SimSiam [8]	0.701	0.778	0.683	0.709
VicReg [23]	0.724	0.783	0.665	0.729
Inter_MoCo [35]	<u>0.741</u>	0.819	0.719	0.744
CAiD [33]	0.707	0.810	0.695	0.731
PCRLv2 [31]	0.726	<u>0.824</u>	<u>0.724</u>	<u>0.746</u>
MLVICX	0.772	0.850	0.748	0.776

The best results are in bold fonts, and the second best are underlined.

only in transfer learning but also maintains its superiority when fine-tuned for a specific downstream task. Notably, MLVICX's exceptional performance highlights its suitability for advancing the field of medical image analysis.

Further, To demonstrate MLVICX's versatility, we performed additional pre-training on the diabetic retinopathy dataset EyePACS [56], which contains 35,126 images. Downstream evaluations are conducted on two retinopathy datasets: MuReD [57], with 2,208 samples across 20 classes, and ODIR [58], [59], with 7,000 retinal images across 8 labels. For both datasets, we considered downstream classification with

limited labeled data, using 10% of samples for training and 20% for validation. The results are summarized in Table IV. The findings in Table IV demonstrate the proposed method's robust performance under both finetuning and frozen evaluations. MLVICX outperforms other methods on MuReD and ODIR datasets, achieving the highest AUC scores. Specifically, for MuReD, MLVICX achieves an AUC score of 0.772 in the frozen setting and 0.850 in the fine-tuning setting, compared to the second-best scores of 0.741 and 0.824, respectively. Similarly, for ODIR, MLVICX achieves an AUC of 0.748 in the frozen setting and 0.776 in the fine-tuning setting, with the second-best scores being 0.724 and 0.746, respectively. The performance

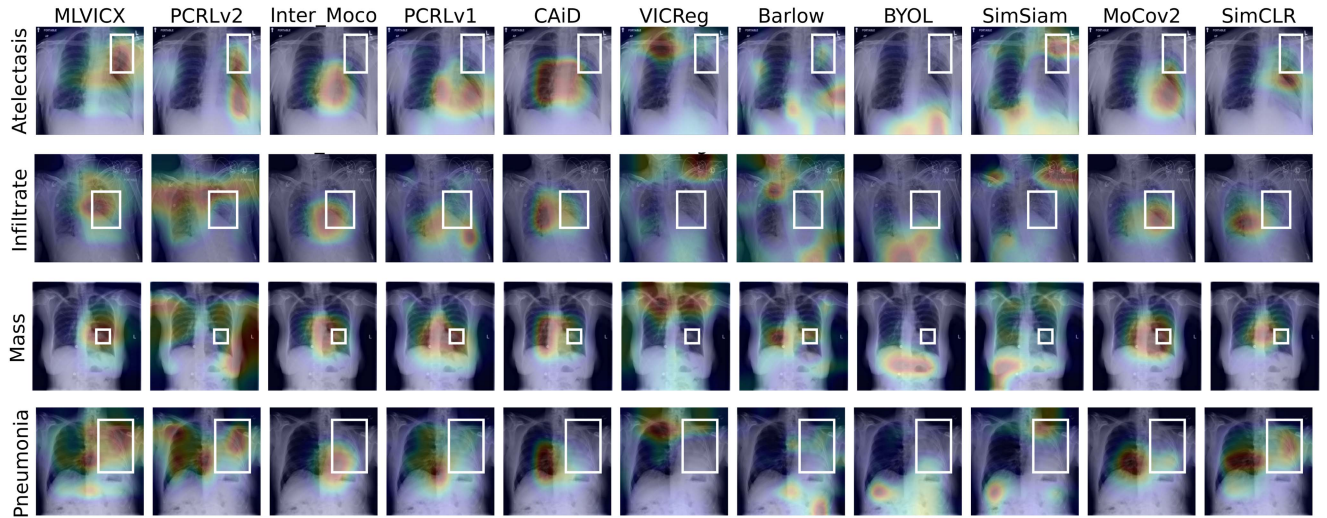


Fig. 2. Diagnostic heatmaps generated by MLVICX and SSL baselines. The heatmaps represent model interpretations of chest X-ray images fine-tuned with 1% of training samples from the NIH dataset.

of the proposed method on retinopathy datasets highlights its adaptability and potential for broader application in different medical image analysis tasks.

A. Statistical Significance Analysis

In this section, we conducted the statistical significance analysis to understand the performance of our proposed model, MLVICX, compared to the best-considered baseline, PCRLv2. We run both models ten times by randomly selecting the subsets, specifically 1%, 10%, and 30% of the samples from the NIH training set, each involving fine-tuning of learned representations. Subsequently, both MLVICX and PCRLv2 undergo fine-tuning of learned representations in each run. p -values are calculated for each data percentage using a paired t-test, considering the results from ten independent runs. With 1% of the data, the p -value is $6.99\text{e-}03$, indicating a statistically significant difference in favor of MLVICX. For 10% and 30% data, the p -value further diminished to $6.27\text{e-}04$ and $2.42\text{e-}07$, respectively. The lower p -value affirms the significant performance advantage of MLVICX over PCRLv2. In all the experiments, the p -value is considerably smaller than 0.01, demonstrating the consistent superiority of MLVICX.

B. Ablation Study

In this section, we conducted an ablation study to examine the impact of the individual loss components and context-bottleneck heads (CBHs) within MLVICX. The model's performance is evaluated across various data percentages on the NIH dataset, both in the fine-tuning and frozen settings, by selectively disabling either the $\mathcal{L}_G$ loss or the $\mathcal{L}_L$ loss while including the context-bottleneck heads. Subsequently, to understand the effect of context-bottleneck heads, we disable them and keep both $\mathcal{L}_G$ loss or the $\mathcal{L}_L$ during pre-training. Table V summarizes the results of this study in terms of AUC. The results in Table V provide insights into the contributions of individual components

TABLE V
RESULTS OF AN ABLATION STUDY CONDUCTED ON MLVICX ACROSS VARIOUS DATA PERCENTAGES ON THE NIH DATASET

Data(%)	Loss		CBHs	1%	10%	30%
	$\mathcal{L}_G$	$\mathcal{L}_L$				
Finetune	×	✓	✓	<u>0.637</u>	<u>0.727</u>	<u>0.759</u>
	✓	×	✓	0.631	0.721	0.751
	✓	✓	×	0.633	0.718	0.747
	✓	✓	✓	0.642	0.733	0.763
Frozen	×	✓	✓	<u>0.589</u>	<u>0.653</u>	<u>0.681</u>
	✓	×	✓	0.581	0.648	0.677
	✓	✓	×	0.580	0.644	0.671
	✓	✓	✓	0.601	0.661	0.691

The study evaluates the impact of individual loss components, $\mathcal{L}_G$ and $\mathcal{L}_L$ and context-bottleneck heads (CBHs), in both fine-tuning and frozen settings. Bold fonts represent the best performance, while underlined text represents the second best.

of the MLVICX model: the global loss component $\mathcal{L}_G$, the local loss component $\mathcal{L}_L$ and the context-bottleneck heads. In the fine-tuning setting with 1% data, the best performance (AUC 0.642) is achieved when both losses and the context-bottleneck heads are included. Excluding CBHs results in a performance drop to 0.633, while excluding $\mathcal{L}_L$ results in an AUC of 0.631, indicating that both $\mathcal{L}_L$ and CBHs contribute positively, especially in low-data scenarios. A similar performance is observed in the frozen setting; with 1% data, the full model achieves the highest AUC (0.601). On excluding any of the $\mathcal{L}_G$ or $\mathcal{L}_L$, a performance drop is observed showcasing the significance of both $\mathcal{L}_G$ and $\mathcal{L}_L$ components in enhancing MLVICX's performance. When $\mathcal{L}_L$ is eliminated, indicating the exclusion of multi-level embeddings, the performance decline is slightly more prominent compared to the removal of $\mathcal{L}_G$. Also, When CBHs are disabled, there is a noticeable drop in the AUC scores in both fine-tuning and frozen settings, demonstrating their importance. This suggests the importance of multi-level embeddings in the model's representation learning process.

C. Qualitative Evaluation

Fig. 2 presents visual interpretations of chest pathology diagnoses for different conditions. We compare the performance of the proposed SSL framework MLVICX with all the considered SSL baselines. The heatmaps correspond to representations, fine-tuned with only 1% of training samples from the NIH dataset. Notably, we observed that MLVICX consistently produces more interpretable diagnostic results when compared to the other methods. Specifically, in the case of pathologies like Mass, Atelectasis, and Infiltrate, it exhibits relatively better performance in comparison with the considered baselines. These pathologies often involve smaller lesion areas, making them more challenging to diagnose accurately. In the case of cardiomegaly, we find that most methods exhibit similar performance; however, MLVICX delivers more consistent and precise predictions. Even in the case of Pneumonia, it outperforms others. This analysis highlights MLVICX's effectiveness in handling a diverse range of pathologies.

VI. CONCLUSION

This work proposes an SSL framework MLVICX that aggregates contextual representations from multiple intermediate layers. We demonstrate the performance of MLVICX on multiple chest X-ray datasets with multiple downstream tasks, including classification and segmentation. MLVICX can leverage limited labeled data while extracting meaningful features from chest X-rays through the combination of multilevel variance-covariance exploration. With its strong performance on various downstream tasks, MLVICX has the potential to become a valuable tool in various chest X-ray analysis applications. MLVICX offers a simple yet effective way to improve the generalization ability of the representations learned in transfer learning and when finetuned for specialized tasks.

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